CompArchView

Operating manual and troubleshooting guide

Credits:

Written and edited by Adam Hamamreh

Version: 1

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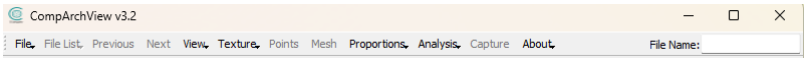
Troubleshooting 17

Note: Due to the active development, not all images will have the same functions or button layouts throughout the manual; however, the instructions still apply to all future versions.

Thank you for installing CompArchView on your computer! Once the installation has been completed, you will see this window.

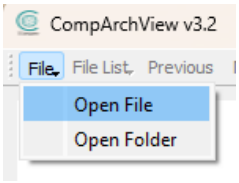


The toolbar, placed on the top portion of the screen, has all the buttons and functions the application can do. Notice that at the far-right corner, there is a File Name field which will display the loaded 3D file's name.



Note: Without a 3D file present, not all functionalities will be available. In addition, some buttons will not be available depending on the file type.

There are two ways you can open a single file in the application. The first way is to click on File and then select Open File. From there, choose the file you wish to open from the file browser.

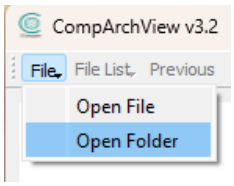


The second way to open a file is by using the Drag and Drop feature. All you must do is drag a supported 3D file to the application window.



Note: The application supports **WRL**, **STL**, **PLY** and **OBJ** files.

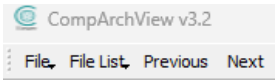
If you have multiple 3D files in a folder, you can select that folder by choosing Open Folder in the File menu.



Note: Alternatively, you can Drag and Drop the whole folder that you whish to open to the application window.

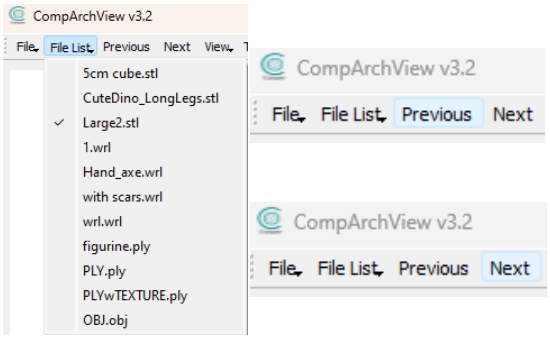


Once you select your folder, notice that a few buttons become available: File list, Previous, and Next.



Note: The Previous and Next buttons may not always be available. They are enabled depending on your position in the File List.

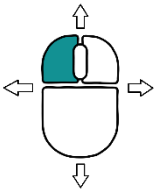
After your first file loads from the folder you selected, you can use the Next and Previous buttons to navigate through the files. It is also possible to click on the File List to select a specific file you wish to see.



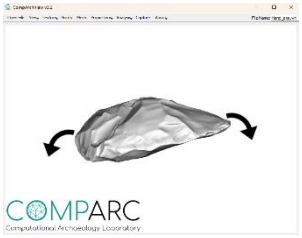
Rotation:

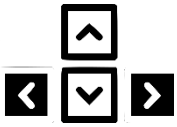
After loading the 3D file, you can rotate the object using your mouse. To do so, press and hold your Left Mouse Button and move your mouse to the direction you want to rotate the object.





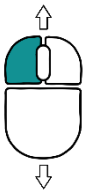
To rotate the object left and right, press and hold the Ctrl key and press the Left or Right arrow keyboard keys.





Another way to rotate is to press and hold the Ctrl key and press and hold the Left Mouse Button while dragging up or down.

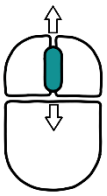




Zoom:

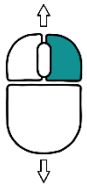
There are multiple ways to change the zoom of the camera. One way is to use the Scroll Wheel. Scrolling up will zoom in, while scrolling down will zoom out.





Another method of zooming is to Right Click and drag your mouse up to zoom in and down to zoom out.





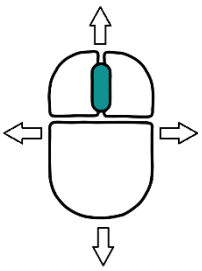
If you are using a touchpad, you can Pinch to Zoom to zoom in and Pinch Out to zoom out.



Alternatively, you can move your mouse cursor to the place you wish to zoom into and press the F key on your keyboard to pan.

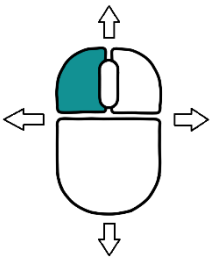
Movement:





There are two ways to move the object. The first way is by pressing and holding the Middle Mouse Button and dragging the object to where you want to place it.



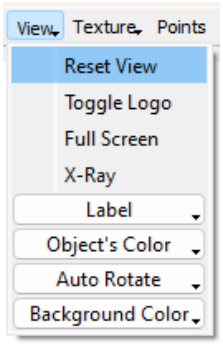


Another way is by pressing and holding the Shift key while Left Clicking on the mouse to move the object.

Note: If you wish to reset the object back to the default position, you can press the R key on your keyboard or press the View button then press Reset View.

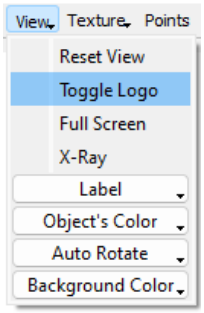
Reset View:

The Reset View button will move the item back to its default position. Alternatively, you can press the R key on your keyboard. This is useful if the object has moved away from the view window.



Toggle Logo:

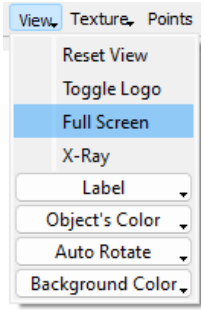
The Toggle Logo button will show or hide the logo, found on the bottom left corner of the window.

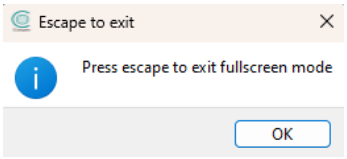




Full Screen:

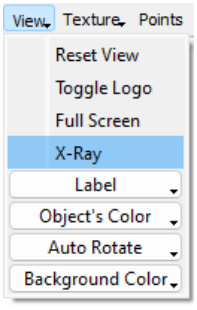
The mode can be enabled by pressing the Full Screen button. To exit the mode, press the Esc key on your keyboard. This mode is recommended to be used with Auto Rotate to display objects. Remember that if you move an object out the view window, you can press R to reset the object's position.





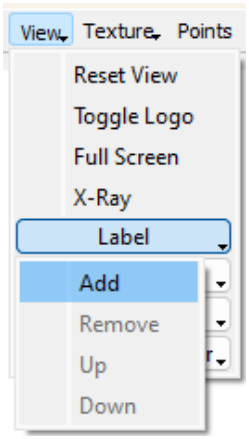
X-Ray:

The X-Ray button sets the opacity of the object to 50%. This feature is useful to visualize the depth of holes or concave features on an object.



Label:

The button will allow you to create a label for the object you have loaded. After selecting the button, you will have four options: Add, Remove, Up and Down. The created label will be placed in the top center position; however, it can be moved up or down.



Once you click on the Add button, a window will come up, prompting you to input the text for the label. For convenience, the input field will be populated with the file's name. However, this can be changed.

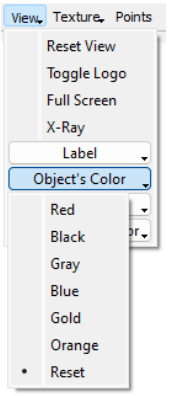
After creating the label, you can move the label Up or Down using the respective buttons.

Once the label is not needed anymore, you can remove it by pressing Remove.

Note: The Remove, Up and Down buttons will be greyed out till a label is created.

Object's Color:

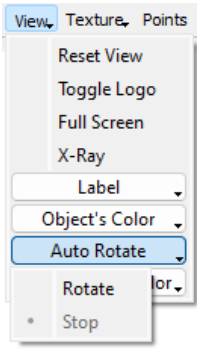
You can change the color of the 3D object by selecting a color from the Object's Color menu. This is useful for revealing hard to see detail or for purely visual needs.



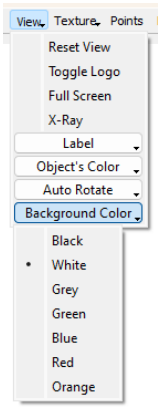
Note: This function is only available for **STL**, **OBJ**, and **WRL** files. If you wish to use it for **PLY** files, you must first select De-Texture PLY in the Texture menu.

Auto Rotate:

The Auto Rotate function automatically rotates the object. This feature is greatly complimented by the Full Screen feature as it allows for objects to be continuously rotated, helping to both visualize and display the artifact.



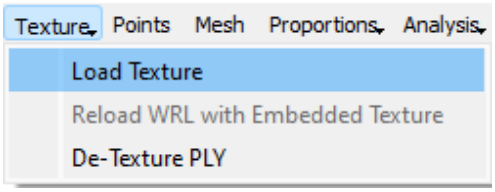
Background Color:



The Background Color menu allows you to select any color you wish from the menu to apply as the background color.

Note: The background of the label and the scale will remain white.

Load Texture:

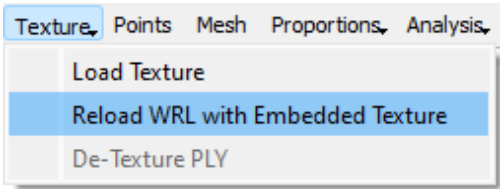


Using the Load Texture button, you can select a **JPG** or a **PNG** texture map to apply to your 3D model.

Alternatively, you can Drag and Drop your texture image on the 3D object.

Note: Only **PLY**, **OBJ**, and **WRL** files support this feature.

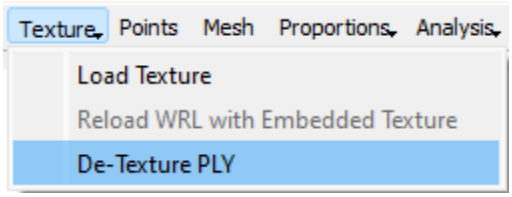
Reload WRL with Embedded Texture:



Sometimes, texture is embedded into the **WRL** file. By default, **WRL** files are not loaded with their texture. If you know that your **WRL** file has embedded texture, press Reload WRL with Embedded Texture.

Note: This option is only available for **WRL** files.

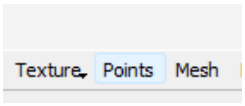
De-Texture PLY:

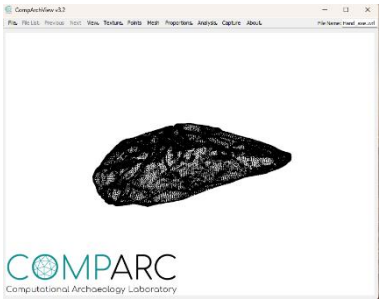


Some **PLY** files might load with unwanted texture. At times, the texture might appear very dark. To fix this issue, select the De-Texture PLY button.

Points:

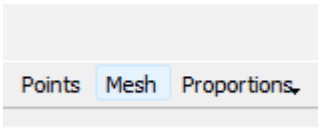
The Points button toggles the points and hides the object's mesh.

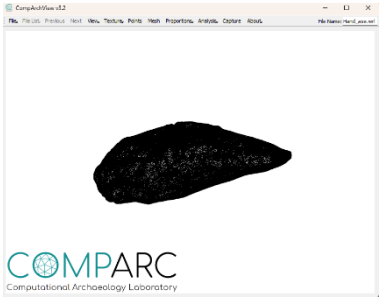




Mesh:

The Mesh button toggles the wireframe of the 3D object.

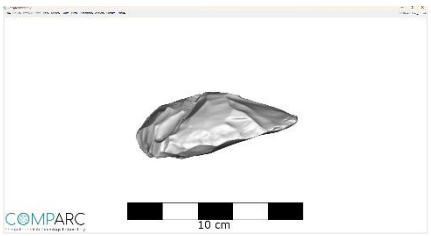




Scales:

The Scales button allows you to select your desired scale length and place under the 3D object. The scale can be adjusted using the Plus and Minus keyboard keys.

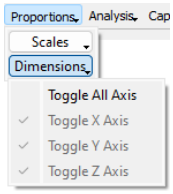


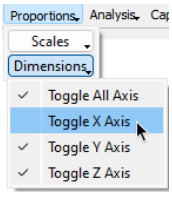


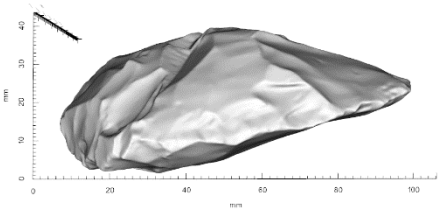
Note: The scale is not automatically calibrated. To calibrate the scale, please go to the next section.

Dimensions:

The Dimensions button allows you to place dimensions on the object's boundaries. You are able to Toggle All Axis dimensions or toggle dimensions for a single axis. The dimensions will always change their angle so they can be in view.







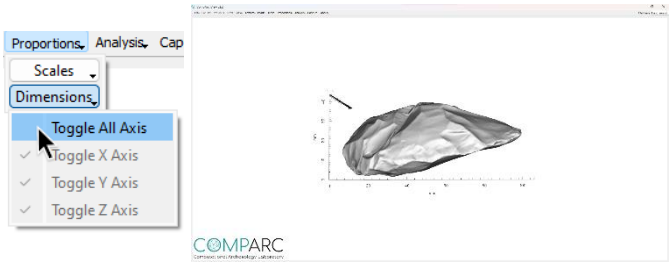
Note: Please make sure that the 3D object is scanned in millimeters, so the dimensions are accurate. Furthermore, the dimensions only display units in millimeters; this cannot be changed.

Calibrating the scale

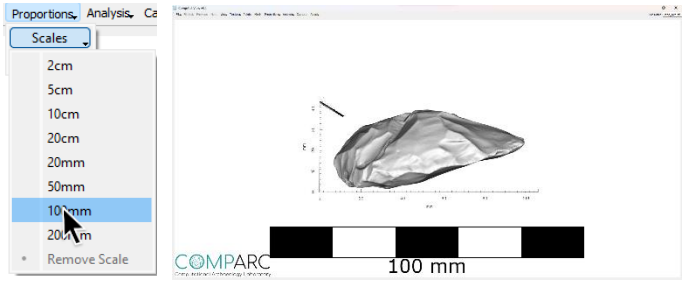
The Scale is not calibrated and will always show the wrong size until it is manually calibrated. This must be done for every 3D object that is loaded.

To calibrate the scale, please follow these steps:

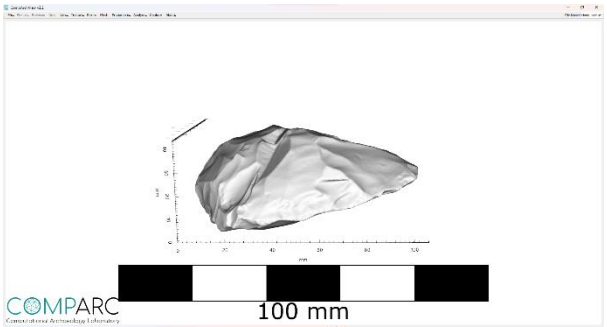
1: Go to the Proportions menu and click on Toggle All Axis to show the dimensions.

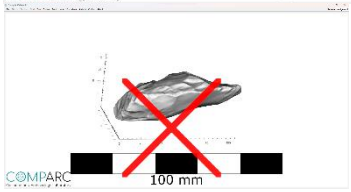


2: Go to the Scales menu and select any scale size that you wish to use. For this example, I will use the 100mm scale.



3: Choose the longest side of the 3D object and line up the dimensions with the scale, like image.

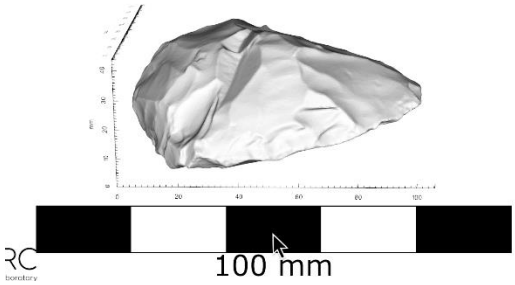




Note: Use the dimension that is closest to the scale to calibrate.

Calibrating the scale 4:

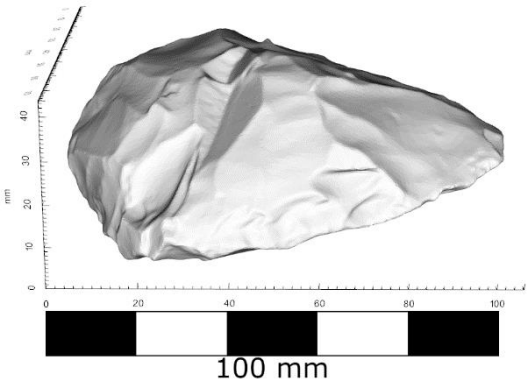
With your mouse, click on the scale and use the Plus (+) and the Minus (-) keyboard keys to adjust the size of the scale. The aim is to get the scale to match the dimensions from the 0 point to the relevant distance. In this example, the scale is 100mm, so the scale will extend from the 0 point to the 100mm point.



Note: Feel free to move the object around during this process. However, remember to click on the scale again to resize it.



Try and get the scale to match the scale as best as you can like the following image:



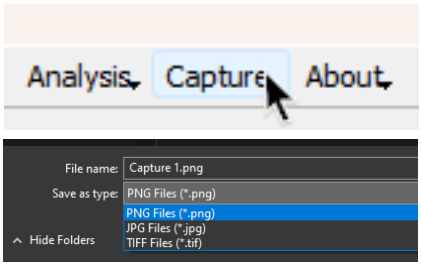
After completing the calibration process, feel free to hide the dimensions and to zoom in and out as you wish. The scale will resize according to the object's zoom level.

Important note 1: Depending on how much you zoom in or out, the Scale might shift from the Dimensions and will show a larger or smaller scale. Thus, making the Scale not accurate. As a result, it is good practice to re-check the Scale size with the Dimensions as shown in step **4** every so often.

Important note 2: The Scale will **not** adjust if you press the Reset View button in the View menu or the R key on your keyboard, which also will reset the view. You will have to re-calibrate the scale if you reset the view.

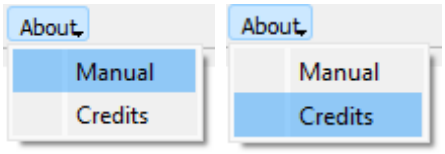
Capture:

The Capture button takes an image of the render window and prompts you to choose the location where you wish to save the image. You are also able to change the format of the image from **PNG** to **JPG** or **TIF**.



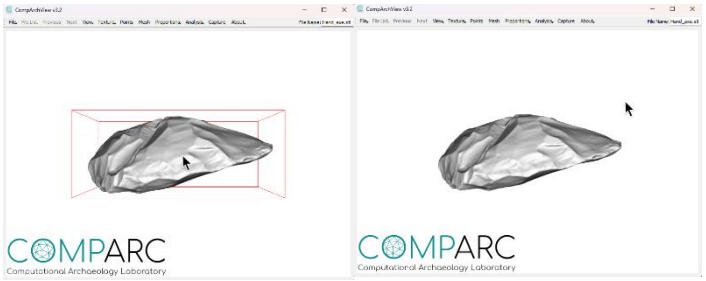
About:

The About menu contains the Manual and the Credits buttons. You can press the Manual button at any time to view this manual for any help that you might need. The Credits button will show a message with the contributor for the program.



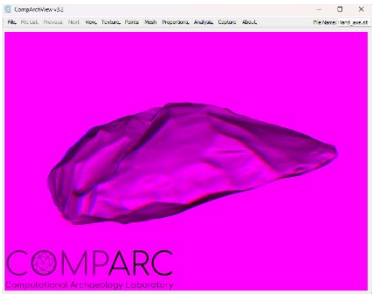
Boundaries:

You can show an object's boundaries by left clicking on the 3D object and pressing the P key on your keyboard. To remove the boundary, move the mouse cursor away from the object and press the P key again.

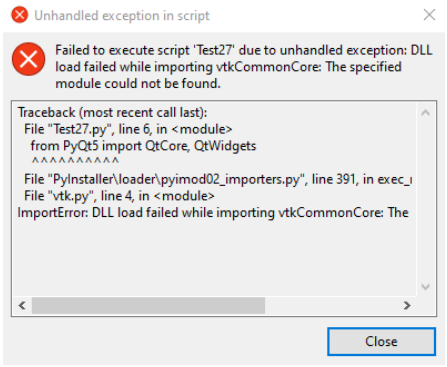
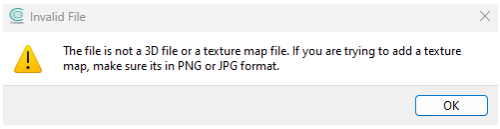
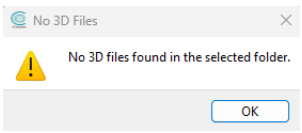


3D effect:

You can toggle the 3D effect by pressing the number 3 on your keyboard. This is meant to be used with red and blue 3D glasses.



Troubleshooting

Issue	Solution
	This error occurs when Visual Studio Redistributable Is **not** installed on your system. Please install the latest version from Microsoft's website that is compatible for your system. Click here to download the x64 bit version of Visual Studio Redistributable. If this error occurs after installation, make sure you do not have any other conflicting versions of Visual Studio Redistributable installed on your system.
	This error occurs if your system does not have its graphics drivers installed. Since there are so many systems and so many drivers, you can search online for your computer's name and the word "graphics drivers". Please make sure that you only install drivers from your computer's manufacturer website.
	This message will appear if you used the Drag and Drop feature and added a file that is not a **PLY**, **STL**, **WRL** or an **OBJ** file. If you did indeed chose such a file, try opening it in another 3D viewer application to see if there is an issue with the file. If you attempted to load a texture map, make sure that it is in the **PNG** or **JPG** format.
	This error occurs if you used the Drag and Drop feature to load a folder that does not contain any 3D files in it. Please check the folder and try again. If this does not work, try restarting the software.